

AS • Edexcel • Physics

 39 mins  4 questions

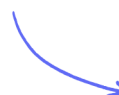
Structured Questions

Motion

Equations of Motion / Motion Graphs / Properties of Motion Graphs / Scalars & Vectors / Resolving Vectors / Finding the Resultant Vector / Projectiles / Forces as Vectors

Medium (2 questions)	/21
Hard (2 questions)	/18
Total Marks	/39

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Medium Questions

- 1 (a) Scientists have been studying a type of spider that can jump up to six times its body length.

The scientists photographed a spider at 0.02s intervals, during a jump. The picture is taken from the photograph and is shown at its actual size.



- i) Deduce whether the images show that the motion in the x-direction is independent of the motion in the y-direction. You should take measurements with a ruler using the cross markings at the centre of gravity of the spider.

(4)

- ii) Show that the initial velocity of the spider at the start of the jump is about 1 m s^{-1} . You should take measurements using the cross marking the centre of gravity of the spider.

(5)

- iii) The spider achieves this jump by extending its two back legs by 3.0 mm.

Calculate the average force the spider exerts in each leg to achieve the jump.

mass of spider = 150 mg

Average force =(3)

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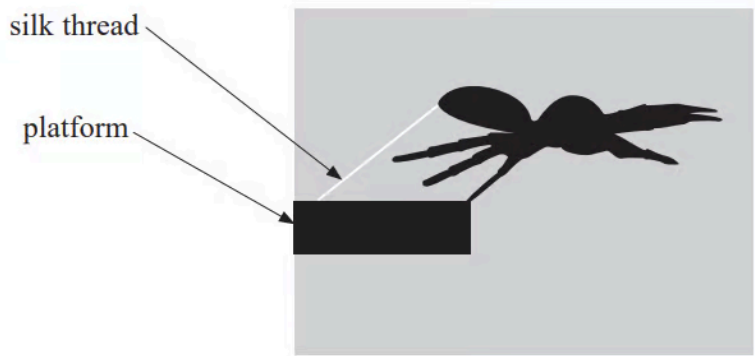
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(12 marks)

(b) Just as the spider starts the jump, it fixes a silk thread to the platform. It is thought that the thread acts as a safety line in case the spider falls.



A student makes the comment:

‘If the silk thread can withstand a tension equal to the weight of the spider then this safety system should work.’

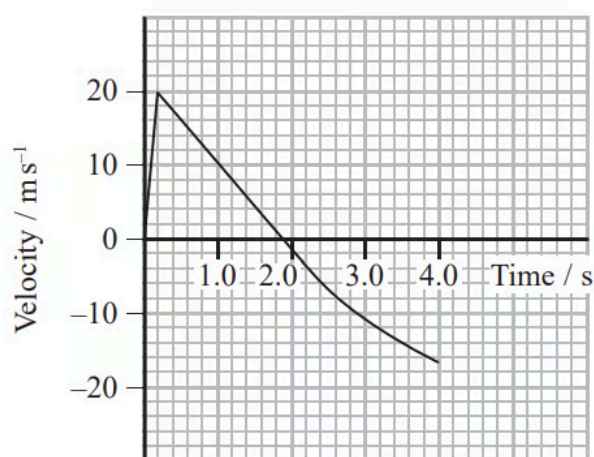
Deduce whether this statement is correct.

(2 marks)

- 2 (a)** A physics class made a toy rocket. A drinks bottle was partially filled with water and inverted over a valve. An air pump delivered air to the bottle until the pressure forced the bottle from the valve and the water was ejected from the bottle at high speed.



A velocity-time graph for the bottle for the first 4 s after take-off is shown.



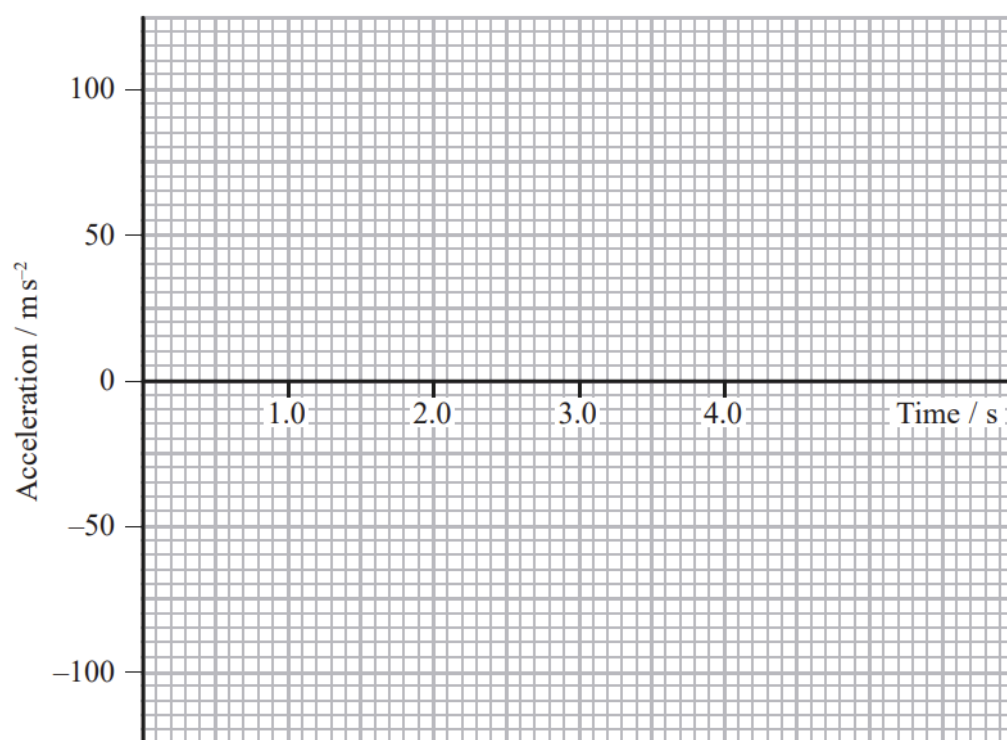
Determine the height to which the rocket travelled.

Height =

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(2 marks)

(b) Sketch the corresponding acceleration-time graph on the axes below.



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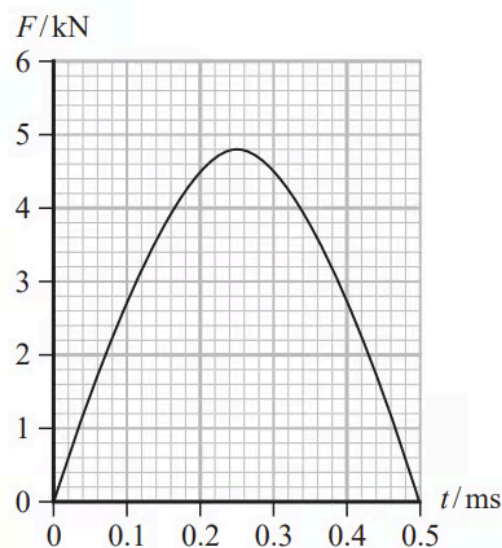
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(5 marks)

Hard Questions

- 1 (a)** In the game of golf a stationary ball is hit by a club. One of the aims of the game is to land the ball on a patch of ground called the green.

The graph shows how the force F exerted by the club on the ball varies with time t as the ball is hit.



State why the area under the graph represents impulse.

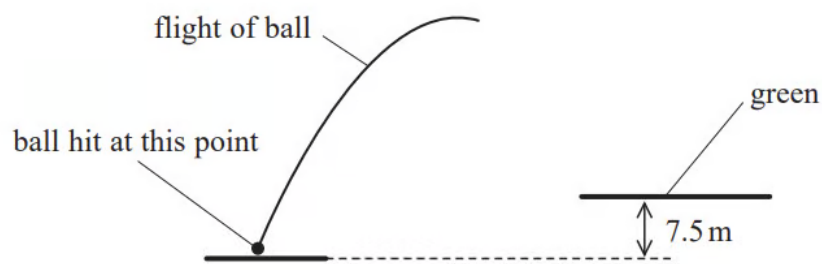
(1 mark)

- (b)** i) Show that the velocity of the ball is about 30 ms^{-1} immediately after it is hit by the club.

mass of ball = 0.046 kg

(3)

ii) The ball has a time of flight of 3.5 s before landing. The green is a vertical distance of 7.5 m above the point at which the ball was hit, as shown. The green is about seventy metres away from where the ball is hit.

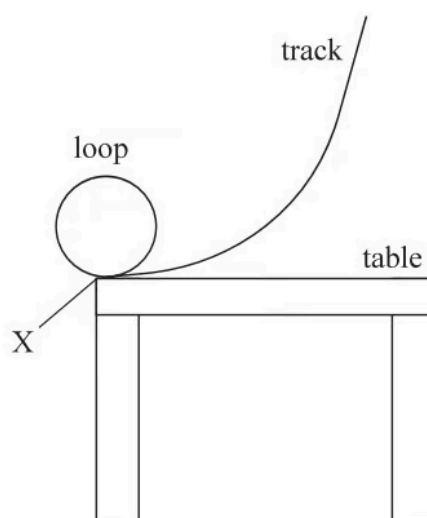


Deduce whether, if air resistance is ignored, the ball could land on the green after a flight time of 3.5 s.

(5)

(8 marks)

2 (a) A track for toy cars can be built with a circular loop as shown.



A toy car is placed on the track at various heights. It travels around the loop before leaving the track horizontally at X.

The loop has radius r and the mass of the toy car is m . It is possible for a toy car to complete the loop without losing contact with the inside of the track.

For this to occur the minimum speed of the toy car at the top of the loop v_{top} is given by

$$v_{top} = \sqrt{gr}$$

Explain why.

(2 marks)

(b) The toy car just completes the loop without losing contact with the track.

Show that the speed of the toy car at the bottom of the loop is about 3 ms^{-1} .

$r = 0.15 \text{ m}$

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(3 marks)

- (c)** The toy car leaves the track at X with a horizontal velocity of 3.0 ms^{-1} .

X is 0.65 m above the floor.

Calculate the horizontal displacement of the car from X when it hits the floor.

Horizontal displacement =

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(4 marks)